

SIMATS ENGINEERING
CO3 ASSESSMENT TOOL 3
MCQ Test

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions

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Assessment Tool Weightage: 20%

QUESTIONS:20

Total Marks:20

1. Which TCP mechanism reduces the congestion window by half upon detecting packet loss via duplicate ACKs?
 - a) Slow Start
 - b) Fast Retransmit
 - c) Fast Recovery
 - d) Tahoe Reset(BL4)(1)
2. A TCP sender has $CWND = 8$ MSS and $SSTHRESH = 16$. After one RTT with no loss, $CWND$ becomes:
 - a) 9 MSS
 - b) 16 MSS
 - c) 12 MSS
 - d) 8 MSS(BL4)(1)
3. Which field in the TCP header is used to implement flow control at the receiver side?
 - a) Sequence number
 - b) Acknowledgement number
 - c) Window size
 - d) Urgent pointer(BL4)(1)
4. Nagle's algorithm causes increased latency primarily because it:
 - a) Disables ACK piggybacking
 - b) Buffers small segments until ACK arrives

20/20 ✓

c) Reduces CWND on every packet

d) Increases SYN backlog

(BL4)(1)

5. In a high-latency satellite link (600 ms RTT), TCP throughput is primarily limited by:

a) MTU size

b) CWND growth rate relative to RTT

c) DNS resolution time

d) UDP competition

(BL4)(1)

6. UDP is preferred over TCP for DNS queries primarily because:

a) UDP supports fragmentation

b) DNS responses are small and connection overhead is unnecessary

c) TCP cannot use port 53

d) UDP has better error correction

(BL4)(1)

7. A UDP-based streaming application experiences 15% packet loss. The best application-layer mitigation is:

a) Increase UDP buffer size

b) Switch to TCP

c) Implement Forward Error Correction (FEC)

d) Reduce frame rate to 1 fps

(BL4)(1)

8. What is the maximum UDP payload size without IP-level fragmentation on a standard Ethernet link (MTU = 1500 bytes)?

a) 1480 bytes

b) 1472 bytes

c) 1460 bytes

d) 1500 bytes

(BL4)(1)

9. QUIC protocol uses UDP as its transport. The key advantage over TCP-based TLS is:

a) Eliminating IP header overhead

b) 0-RTT or 1-RTT connection setup

c) Removing encryption

d) Native multicast support

(BL4)(1)

10. HTTP/1.1 persistent connections reduce latency by:

a) Enabling UDP fallback

b) Reusing the same TCP connection for multiple requests

c) Compressing HTTP headers

d) Sending requests without waiting for DNS

(BL4)(1)

11. Head-of-line blocking in HTTP/1.1 occurs when:

- a) The DNS server is slow
- b) A response blocks subsequent responses on the same connection
- c) TCP CWND drops to zero
- d) The server rejects pipelined requests

(BL4)(1)

12. HTTP/2 multiplexing solves HOL blocking at the HTTP layer but not at the:

- a) Application layer
- b) DNS layer
- c) TCP layer
- d) IP layer

(BL4)(1)

13. Which HTTP status code indicates that a resource has permanently moved and browsers should update bookmarks?

- a) 301
- b) 302
- c) 304
- d) 307

(BL4)(1)

14. In HTTP caching, the Cache-Control: no-cache directive means:

- a) Never cache this resource
- b) Validate with the server before using a cached copy
- c) Cache for 0 seconds
- d) Disable all caching headers

(BL4)(1)

15. Which DNS record type maps a domain name to an IPv6 address?

- a) A
- b) CNAME
- c) AAAA
- d) MX

(BL4)(1)

16. A DNS resolver switches from UDP to TCP when:

- a) The query type is MX
- b) The response is truncated (TC bit set)
- c) TTL exceeds 3600 s
- d) The authoritative server is unreachable

(BL4)(1)

17. Which attack exploits the UDP-based DNS protocol by sending forged responses with a matching transaction ID?

- a) ARP poisoning
- b) DNS cache poisoning
- c) SYN flooding
- d) BGP hijacking

(BL4)(1)

18. Setting a very low TTL (e.g., 30 s) for a DNS record primarily increases:
- Cache hit ratio
 - Recursive resolver query load
 - DNSSEC signature validity
 - Zone transfer frequency
- (BL4)(1)
19. A user browses to a new website. The correct sequence of protocol interactions for the first byte of the page is:
- HTTP → TCP → DNS
 - DNS → TCP handshake → HTTP GET
 - TCP → DNS → HTTP
 - DNS → HTTP → TCP
- (BL4)(1)
20. Under 20% packet loss, which protocol combination gives the best performance for a real-time multiplayer game?
- TCP for game state, HTTP for assets
 - UDP with application-layer sequencing for game state, HTTP/3 for assets
 - HTTP/2 for all traffic
 - TCP with Nagle enabled for all traffic
- (BL4)(1)

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers

Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

1. Which TCP mechanism reduces the congestion window by half upon detecting packet loss via duplicate ACKs?

Answer: c) Fast Recovery

Justification: Fast Retransmit triggers on 3 duplicate ACKs, but Fast Recovery is the mechanism that halves the congestion window.

2. A TCP sender has $CWND = 8$ MSS and $SSTHRESH = 16$. After one RTT with no loss, $CWND$ becomes:

Answer: b) 16 MSS

Justification: Since $CWND (8) < SSTHRESH (16)$, the sender is in slow start, where $CWND$ doubles each RTT with no loss: $8 \rightarrow 16$.

3. Which field in the TCP header is used to implement flow control at the receiver side?

Answer: c) Window size

Justification: The receiver advertises available buffer space via the Window Size field.

4. Nagle's algorithm causes increased latency primarily because it:

Answer: b) Buffers small segments until ACK arrives

Justification: Nagle's algorithm holds small segments until a pending ACK is received before sending, adding delay.

5. In a high-latency satellite link (600 ms RTT), TCP throughput is primarily limited by:

Answer: b) $CWND$ growth rate relative to RTT

Justification: Throughput is capped by how fast $CWND$ can grow relative to RTT (the bandwidth-delay product effect).

6. UDP is preferred over TCP for DNS queries primarily because:

Answer: b) DNS responses are small and connection overhead is unnecessary

Justification: Small query/response sizes make TCP's connection setup overhead unnecessary.

7. A UDP-based streaming application experiences 15% packet loss. The best application-layer mitigation is:

Answer: c) Implement Forward Error Correction (FEC)

Justification: FEC lets the receiver reconstruct lost data without needing retransmission, ideal for lossy real-time streams.

8. What is the maximum UDP payload size without IP-level fragmentation on a standard Ethernet link (MTU = 1500 bytes)?

Answer: b) 1472 bytes

Justification: $1500 \text{ (MTU)} - 20 \text{ (IP header)} - 8 \text{ (UDP header)} = 1472 \text{ bytes}$.

9. QUIC protocol uses UDP as its transport. The key advantage over TCP-based TLS is:

Answer: b) 0-RTT or 1-RTT connection setup

Justification: QUIC combines transport and TLS handshakes to reduce connection setup time.

10. HTTP/1.1 persistent connections reduce latency by:

Answer: b) Reusing the same TCP connection for multiple requests

Justification: Avoiding repeated TCP handshakes for each request reduces overhead.

11. Head-of-line blocking in HTTP/1.1 occurs when:

Answer: b) A response blocks subsequent responses on the same connection

Justification: Responses must be returned in order, so a slow response delays all queued ones.

12. HTTP/2 multiplexing solves HOL blocking at the HTTP layer but not at the:

Answer: c) TCP layer

Justification: All HTTP/2 streams share one TCP connection, so a lost TCP segment still blocks everything.

13. Which HTTP status code indicates that a resource has permanently moved and browsers should update bookmarks?

Answer: a) 301

Justification: 301 Moved Permanently signals a permanent redirect.

14. In HTTP caching, the Cache-Control: no-cache directive means:

Answer: b) Validate with the server before using a cached copy

Justification: no-cache does not disable caching; it requires revalidation before use.

15. Which DNS record type maps a domain name to an IPv6 address?

Answer: c) AAAA

Justification: AAAA records map hostnames to IPv6 addresses (A records are for IPv4).

16. A DNS resolver switches from UDP to TCP when:

Answer: b) The response is truncated (TC bit set)

Justification: The TC bit signals the response didn't fit in a UDP datagram, prompting a TCP retry.

17. Which attack exploits the UDP-based DNS protocol by sending forged responses with a matching transaction ID?

Answer: b) DNS cache poisoning

Justification: Attackers forge responses with a guessed/matching transaction ID to poison resolver caches.

18. Setting a very low TTL (e.g., 30 s) for a DNS record primarily increases:

Answer: b) Recursive resolver query load

Justification: Low TTL forces frequent re-resolution, increasing load on the recursive resolver.

19. A user browses to a new website. The correct sequence of protocol interactions for the first byte of the page is:

Answer: b) DNS → TCP handshake → HTTP GET

Justification: Name resolution must happen before a TCP connection can be opened, before HTTP can be used.

20. Under 20% packet loss, which protocol combination gives the best performance for a real-time multiplayer game?

Answer: b) UDP with application-layer sequencing for game state, HTTP/3 for assets

Justification: UDP avoids TCP's HOL blocking for game state; HTTP/3 (QUIC/UDP) handles asset loading efficiently.

Quick-Reference Answer Sheet

Q	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans	c	b	c	b	b	b	c	b	b	b	b	c	a	b	c	b	b	b	b	b